

Functional Annotation of GlgE (gene *glgE* / PP_4060; UniProt Q88FM9) in *Pseudomonas putida* KT2440

1. Summary (answer to the research question)

GlgE (Q88FM9, locus PP_4060) is α -maltose-1-phosphate:(1 $\rightarrow$ 4)- α -D-glucan 4- α -D-maltosyltransferase (GMPMT; EC 2.4.99.16), a cytoplasmic glycoside-hydrolase family 13 (GH13_3) enzyme. Its primary function is to **elongate a linear α -1,4-glucan chain by transferring maltosyl (two-glucose) units from the phosphosugar donor α -maltose-1-phosphate (M1P) onto the non-reducing end of a maltooligosaccharide/ α -glucan acceptor**. It is the committed polymerizing step of the cytoplasmic **TreS–Mak(Pep2)–GlgE–GlgB pathway**, an ADP-glucose-independent route that converts trehalose/maltose into a branched, glycogen-like α -glucan used for carbon/energy storage. In *P. putida* KT2440 this pathway is genomically complete and clustered (its M1P-supplying maltokinase is immediately adjacent to *glgE*), and — because the classical ADP-glucose enzyme GlgC is absent — it is most likely the organism's **primary** α -glucan biosynthetic route.

The gene identity is **confirmed**: the symbol *glgE*, the EC number, the GH13/GlgE family and domain set (IPR026585 GlgE; IPR006047 GH13 catalytic; IPR049171 GLGE_C; IPR021828 GlgE_dom_N/S) all match the well-characterized GlgE maltosyltransferase. There is no ambiguity with an unrelated gene of the same symbol.

2. Gene/Protein Identity Verification

Attribute	Provided (UniProt/HAMAP MF_02124)	Literature consensus	Match
Enzyme name	GMPMT, maltosyltransferase	α -maltose-1-P: α -1,4-glucan maltosyltransferase	☐
EC number	2.4.99.16	2.4.99.16 (Syson et al. 2014, P 24689960)	☐
Family	GH13, GlgE	CAZy GH13_3 (Syson et al. 2011/2014)	☐
Domains	GH13 cat., GlgE, GLGE_C, GlgE_N/S	5-domain GlgE architecture (P 21914799)	☐
Organism	<i>P. putida</i> KT2440	ortholog; function assigned by orthology	☐ (see §6)

Caveat & strength of assignment: The deep biochemical/structural characterization of GlgE comes from *Streptomyces coelicolor*, *Mycobacterium tuberculosis*, *M. thermoresistibile*, and *S. venezuelae*; the *P. putida* protein itself has not been enzymatically studied. Nonetheless the functional assignment is supported by **five convergent lines of evidence**: (i) the HAMAP rule MF_02124 + curated UniProt reaction/EC; (ii) strong orthology to biochemically/structurally solved GlgE; (iii) conservation of the exact GH13 catalytic triad (Asp391/Glu420/Asp478) in the *P. putida* sequence; (iv) a complete, operonically clustered TreS–Mak–GlgE–GlgB pathway with the M1P-supplying maltokinase adjacent to *glgE* and GlgC absent; and (v) organism-level evidence that *P. putida* makes and rapidly mobilizes α -1,4-glucan (glycogen). This is a high-confidence assignment despite the absence of a direct enzyme assay.

3. Primary Function and Substrate Specificity

- **Reaction:** α -maltose-1-phosphate + [(1 $\rightarrow$ 4)- α -D-glucosyl]_n $\rightarrow$ phosphate + [(1 $\rightarrow$ 4)- α -D-glucosyl]_{n+2}. GlgE was directly identified as "a maltosyltransferase that uses maltose 1-phosphate" (Kalscheuer et al. 2010,

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and formally defined as "an α -maltose 1-phosphate:(1 $\rightarrow$ 4)- α -D-glucan 4- α -D-maltosyltransferase of the CAZy glycoside hydrolase 13_3 family... the defining enzyme of a bacterial α -glucan biosynthetic pathway" (Syson et al. 2014,

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- **Donor specificity: α -maltose-1-phosphate** — a phosphosugar, not ADP-/UDP-glucose. This distinguishes the GlgE route from the classical GlgC/GlgA glycogen pathway.
- **Acceptor specificity:** linear α -1,4-linked maltooligosaccharides/ α -glucan. Ligand-bound structures mapped acceptor subsites +1 to +6; "the sugar residues in the acceptor subsites +1 to +5 are oriented such that they disfavor the binding of malto-oligosaccharides that bear branches at their 6-positions, consistent with the known acceptor chain specificity of GlgE" (Syson et al. 2016,

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GlgE therefore extends **unbranched** stretches, leaving α -1,6 branching to GlgB.

P. putida-specific confirmation (this work). UniProt Q88FM9 (661 aa) itself is annotated with the reaction " α -maltose 1-phosphate + [(1 $\rightarrow$ 4)- α -D-glucosyl]_n = [(1 $\rightarrow$ 4)- α -D-glucosyl]_{n+2} + phosphate (EC 2.4.99.16)" and function "Maltosyltransferase that uses maltose 1-phosphate (M1P) as the sugar donor to elongate linear or branched α -(1 $\rightarrow$ 4)-glucans... together with TreS, Mak and GlgB," localized to a GH13 catalytic domain (residues 211–556), homodimeric. This directly ports the reaction/substrate specificity to the *P. putida* protein.

4. Catalytic Mechanism (evidence from structure + biochemistry)

- **α -retaining double-displacement.** GlgE "catalyzes the α -retaining transfer of maltosyl units from α -maltose 1-phosphate to maltooligosaccharides and is predicted to use a double-displacement mechanism" (Syson et al. 2014,

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- **Covalent intermediate.** A trapped β -2-deoxy-2-fluoromaltosyl-enzyme intermediate and mass spectrometry showed the catalytic nucleophile is an aspartate: "The covalent modification of Asp394 was confirmed using mass spectrometry"

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Glu423 acts as the general acid/base. This is classic GH13 (α -amylase clan) chemistry.

- **Two half-reactions:** (i) maltosyl transfer from M1P to the Asp nucleophile with release of inorganic phosphate; (ii) transfer of the maltosyl group to the 4-OH of the acceptor's non-reducing end, regenerating the α -anomer.
- **Catalytic residues are conserved in the *P. putida* enzyme (this work).** Direct inspection of Q88FM9 confirms the canonical GH13 Asp-Glu-Asp triad — **Asp391** (nucleophile, in the β 4 "RVDNP" motif), **Glu420** (general acid/base, "AE" motif), and **Asp478** (transition-state stabilizer, "TPD" motif) — positionally equivalent to the experimentally validated *S. coelicolor* Asp394/Glu423. UniProt independently annotates residues 391 (nucleophile), 420 (proton donor) and 478 (transition-state stabilizer). This provides sequence-level evidence (beyond generic orthology) that the *P. putida* protein is catalytically competent with the same chemistry.

5. Structure and Localization

- **Quaternary/fold:** "The *S. coelicolor* enzyme forms a homodimer with each subunit comprising five domains, including a core catalytic α -amylase-type domain A with a $(\beta/\alpha)_8$ fold" (Syson et al. 2011,

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it additionally contains an S-domain helix bundle not previously seen in GH13. High-resolution structures of *M. thermoresistibile* (1.96 Å;

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and *M. tuberculosis*

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confirm the conserved architecture.

- **Localization: cytoplasmic (cytosolic).** "GlgE is a bacterial maltosyltransferase that catalyzes the elongation of a **cytosolic**, branched α -glucan" (Lindenberger et al. 2015,

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α -glucan is "exclusively assembled **intracellularly**" (Koliwer-Brandl et al. 2016,

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In mycobacteria a fraction of the polymer is subsequently exported to form a capsule, but the GlgE reaction itself occurs in the cytoplasm.

- **P. putida-specific localization evidence (this work).** UniProt Q88FM9 carries **no signal peptide and no transmembrane region** (features limited to catalytic Domain/Active-site/Binding-site; keywords "Carbohydrate metabolism, Glycosyltransferase, Transferase"), consistent with a **soluble cytoplasmic** enzyme — matching the cytosolic localization of characterized orthologs.

6. Pathway Context and Physiological Role

- **Pathway:** the four-step cytoplasmic GlgE pathway — "a new pathway from trehalose to alpha-glucan... comprising four enzymatic steps mediated by TreS, Pep2, GlgE" and GlgB (

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TreS interconverts trehalose $\leftrightarrow$ α -maltose; maltose kinase Pep2/Mak makes M1P; GlgE polymerizes; GlgB branches (α -1,6). GlgE's substrate M1P is the pathway's key toxic intermediate.

- **Necessity/sufficiency:** in *S. venezuelae*, a $\Delta glgE$ mutant "accumulated α -maltose 1-phosphate and maltose but no α -glucan. Therefore, the GlgE pathway is necessary and sufficient for polymer biosynthesis" (Miah et al. 2016,

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- **Toxic intermediate / drug-target biology (in pathogens):** in *M. tuberculosis*, GlgE inactivation causes self-poisoning through M1P accumulation and rapid death, validating GlgE as an anti-TB target

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This is relevant mechanistically (explains why the M1P $\rightarrow$ polymer step is essential) but is a pathogen-specific consequence, not a role in *P. putida*.

- **Genomic pathway context in *P. putida* KT2440 (this work).** Mapping the KT2440 proteome (taxid 160488) places *glgE* (PP_4060) in a contiguous α -glucan/glycogen gene cluster: *glgA* glycogen synthase (PP_4050), *treZ* (PP_4051), *malQ* amyloamylase/4- α -glucanotransferase (PP_4052), *treY* (PP_4053), *glgX* debranching enzyme (PP_4055), *glgB* branching enzyme (PP_4058), maltokinase *treSB* (PP_4059) and *glgE* (PP_4060); *glgP* glucan phosphorylase (PP_5041) and trehalose synthase *treSA* (PP_2918) lie elsewhere. Crucially, the gene immediately upstream of *glgE* — maltokinase *treSB*/

PP_4059 (Q88FN0) — carries the exact donor-supplying reaction "**D-maltose + ATP = α -maltose-1-phosphate + ADP + H⁺**" (EC 2.7.1.175), i.e., the Mak/Pep2 step that feeds M1P to GlgE, plus a trehalose-synthase (maltose $\leftrightarrow$ trehalose, EC 5.4.99.16) activity. Thus *P. putida* has a **complete, operonically clustered TreS–Mak–GlgE–GlgB pathway**, with GlgE's donor-generating enzyme encoded right next to it — strong genomic evidence the pathway operates in vivo.

- **GlgC is absent (two databases) → GlgE route is primary.** No glucose-1-phosphate adenylyltransferase / **GlgC (EC 2.7.7.27, KEGG K00975)** exists in KT2440 — confirmed independently by UniProt (gene/protein/EC search) and KEGG (org "ppu" returns no glgC gene). The resident **GlgA (PP_4050, Q88FN9)** is a *canonical* ADP-glucose-dependent glycogen synthase (EC 2.4.1.21, GT1 family) that strictly requires ADP-glucose; with GlgC absent, this classical route cannot run conventionally. Therefore the **M1P-dependent GlgE pathway is the primary α -glucan biosynthetic route** in *P. putida*. (This corrects the earlier assumption of an intact classical GlgC–GlgA route.)
- **The M1P supplier is a bifunctional TreS–maltokinase (this work).** PP_4059 (Q88FN0), immediately adjacent to *glgE*, is annotated with **both** trehalose-synthase (D-maltose $\rightleftharpoons$ trehalose, EC 5.4.99.16) **and** maltokinase (D-maltose + ATP $\rightarrow$ α -maltose-1-phosphate + ADP, EC 2.7.1.175) activities — a fused TreS–Mak enzyme that both makes maltose from trehalose and phosphorylates it to GlgE's donor, tightly coupling substrate supply to GlgE.
- **Role in *P. putida*:** the physiological output is an intracellular **glycogen-like α -1,4/ α -1,6-glucan** for **carbon/energy storage**. *P. putida* KT2440 "access[es] cellular PHA, amino acids and glycogen in few seconds under glucose starvation to obtain ATP" (Ankenbauer et al. 2020,

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and a *P. putida* polysaccharide was shown to be a " $\rightarrow$ 4)- α -d-Glcp-(1 $\rightarrow$ glucan (bacterial glycogen)" (Zdorovenko et al. 2018,

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7. Supported and Refuted Hypotheses

Supported - H1: GlgE is a maltosyltransferase using M1P as donor (EC 2.4.99.16). □

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- H2: Mechanism is α -retaining double-displacement via a covalent maltosyl-Asp intermediate. □

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- H3: GlgE extends linear α -1,4 chains; branching disfavored at acceptor site. □

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- H4: Cytoplasmic homodimeric GH13 enzyme. □

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- H5: Acts in the TreS-Pep2-GlgE-GlgB trehalose→ α -glucan pathway. □

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Refuted / not applicable - That GlgE uses ADP-glucose/UDP-glucose as donor — refuted; the donor is the phosphosugar M1P. - That the *glgE* symbol is ambiguous — refuted; all identifiers converge on GlgE maltosyltransferase. - Pathogen "self-poisoning/essentiality" is documented in mycobacteria and cannot be assumed to hold in *P. putida* (which is non-pathogenic); note, however, that *P. putida* KT2440 **lacks GlgC**, so the GlgE pathway is not obviously redundant here and may be the main α -glucan route (its essentiality/dispensability under given conditions remains to be tested experimentally).

8. Limitations and Future Directions

- No direct enzymatic, structural, or genetic characterization of the *P. putida* GlgE (PP_4060) protein exists; the functional assignment rests on HAMAP + orthology + conserved catalytic

triad + genomic pathway completeness + organism-level glycogen physiology (all consistent).

- The M1P supply route is now genomically resolved: an adjacent maltokinase (treSB/PP_4059, EC 2.7.1.175) and a trehalose synthase (treSA/PP_2918) are present, and GlgC (EC 2.7.7.27) is absent — implying the GlgE route is primary. However, the *relative in vivo flux* and whether GlgA contributes M1P (as in mycobacteria) remain to be measured. Future work: $\Delta glgE/\Delta treSB$ mutants, M1P quantification, and glycogen assays under carbon fluctuation.
- Direct proof of cytoplasmic localization and of the exact α -glucan product structure in *P. putida* (vs. inference from orthologs and a related *P. putida* strain) would require biochemical/cell-fractionation studies.

Key References

- Kalscheuer et al. 2010, *Nat Chem Biol* —

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- Syson et al. 2011, *J Biol Chem* (first GlgE structure) —

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- Syson et al. 2014, *J Biol Chem* (mechanism, EC, covalent intermediate) —

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- Lindenberger et al. 2015 (Mtb GlgE structures, cytosolic α -glucan) —

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- Mendes et al. 2015 (M. thermoresistibile structure) —

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- Syson et al. 2016 (acceptor subsites/specificity) —

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- Koliwer-Brandl et al. 2016 (metabolic network, intracellular assembly) —

[P 27513637](#)

- Miah et al. 2016 (Δ glgE necessity/sufficiency) —

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- Ankenbauer et al. 2020 (*P. putida* glycogen energy buffer) —

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- Zdorovenko et al. 2018 (*P. putida* α -glucan/glycogen) —

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